

Chapter 1

Section 1 - Basic Usage

1.1 Section 1.1 - Using Commands and Theorems

Everything here will just be an example

1.1.1 Propositions

Definition 1 (Proposition)

A **proposition** is a statement that is either

Example 1 (Examples with Propositions)

Determine if the following are propositions

1. It's all coming back to me now

Solution:

yes

2. Hello.

Solution:

No

3. $x + 3 = 10$

Solution:

No, since we don't know x

4. In a right triangle, $a^2 + b^2 = c^2$

Solution:

No, since a, b, c are not assigned anything

5. In a right triangle having legs of length a and b , and hypotenuse of c , then $a^2 + b^2 = c^2$

Solution:

Yes, since a, b, c are properly specified

6. There are 75 people in this room or the sky is purple

Solution:

Yes

Note

7 is a **compound proposition**

Note

We denote separate propositions as: p, q, r

1.1.2 Logical Operations (connectives)

Name	Notation	Phrasing	Truth Value
Disjunction (or)	$p \vee q$	"p or q"	True when p, q or both are TRUE
Conjunction (and)	$p \wedge q$	"p and q"	True when both are TRUE
Negation (not)	$\neg p$	"not p"	The opposite truth value of p
Exclusive Or (XOR)	$p \oplus q$	"p or q, but not both"	TRUE when only one is TRUE